

TOC (CS302)

Unit-wise Syllabus Breakdown

Unit 1 – DFA & NFA

Study of finite automata, state transitions, DFA, NFA, and ϵ -NFA.

Unit 2 – Regular Languages

Regular expressions, regular languages, closure properties, and pumping lemma.

Unit 3 – Context-Free Grammars

CFGs, derivations, parse trees, ambiguity, and CNF.

Unit 4 – Pushdown Automata

Stack-based automata, PDA construction, and PDA-CFG equivalence.

Unit 5 – Turing Machines & Decidability

Turing Machines, decidability, recursive languages, and the halting problem.